

Trainings ▾

Engine Noise Diagnostic Procedures - General Information

When diagnosing engine noise problems, make sure that noises caused by accessories, such as the air compressor and power takeoff, are **not** mistaken for engine noises. Remove the accessory drive belts to eliminate noise caused by these units. Noise will also travel to other metal parts **not** related to the problem. The use of a stethoscope can help locate an engine noise.

Engine noises heard at the crankshaft speed, engine rpm, are noises related to the crankshaft, rods, pistons, and piston pins. Noises heard at the camshaft speed, one-half of the engine rpm, are related to the valve train. A hand-held digital tachometer can help to determine if the noise is related to components operating at the crankshaft or camshaft speed.

There is **not** a definite rule or test that will positively determine the source of a noise complaint.

Engine driven components and accessories, such as gear-driven fan clutches, hydraulic pumps, belt-driven alternators, air-conditioning compressors, and turbochargers can contribute to engine noise. Use the following information as a guide to diagnosing engine noise.

(Refer to Engine Noise Excessive - Main Bearing symptom tree)

The noise caused by a loose main bearing is a loud dull knock heard when the engine is pulling a load. If all main bearings are loose, a loud clatter will be heard. The knock is heard regularly every other revolution. The noise is the loudest when the engine is lugging or under heavy load. The knock is duller than a connecting rod noise. Low oil pressure can also accompany this condition.

If the bearing is **not** loose enough to produce a knock by itself, the bearing can knock if the oil is too thin, or if there is no oil at the bearing.

An irregular noise can indicate worn crankshaft thrust bearings.

An intermittent sharp knock indicates excessive crankshaft end clearance. Repeated clutch disengagements can cause a change in the noise.

(Refer to Engine Noise Excessive - Connecting Rod symptom tree)

Connecting rods with excessive clearance knock at all engine speeds, and under both idle and load conditions. When the bearings begin to become loose, the noise can be confused with piston slap or loose piston pins. The noise increases in volume with engine speed. Low oil pressure can also accompany this condition.

(Refer to Engine Noise Excessive - Piston symptom tree)

It is difficult to tell the difference between piston pin, connecting rod, and piston noise. A loose piston pin causes a loud double knock which is usually heard when the engine is idling. When the injector to this cylinder is cut out, a noticeable change will be heard in the sound of the knocking noise. However, on some engines the knock becomes more noticeable when the vehicle is operated on the road at steady speed condition.

Driveability/Low Power/Excessive Fuel Consumption - Checklist

Driveability is a term which in general describes vehicle performance on the road. Driveability problems for an engine can be caused by several different factors. Some of the factors are engine related and some are **not**.

Before troubleshooting, it is important to determine the exact complaint and whether the engine has a real driveability problem or if it simply does **not** meet driver expectations. The Driveability-Low Power Customer Complaint Form is a valuable list of questions that **must** be used to assist the service technician in determining what type of driveability problem the vehicle is experiencing. Complete the checklist before troubleshooting the problem. The form can be found at the end of this section. If an engine is performing to factory specifications but does **not** meet the customer's expectations, it **must** be explained to the customer that nothing is wrong with the vehicle and why.

The troubleshooting symptom charts have been set up to divide driveability problems into two different symptoms: Engine Power Output Low and Engine Acceleration or Response Poor.

Low power is a term that is used in the field to describe many different performance problems. However, in this manual low power is defined as the inability of the engine to produce the power necessary to move the vehicle at a speed that can be reasonably expected under the given conditions of load, grade, wind, and so on. Low power is usually caused by the lack of fuel flow which can be caused by any of the following factors:

- Lack of full travel of the throttle pedal
- Failed boost sensor
- Excessive fuel inlet, intake, exhaust, or drain line restriction
- Loose fuel pump suction lines.

Low power is **not** the inability of the vehicle to accelerate satisfactorily from a stop or the bottom of a grade. See the performance tree Engine Power Output Low for the proper procedures to locate and correct a low power problem. The chart starts off with basic items which can cause lower power.

Poor acceleration or response is described in this manual as the inability of the vehicle to accelerate satisfactorily from a stop or from the bottom of a grade. It can also be the lag in acceleration during an attempt to pass or overtake another vehicle at conditions less than rated speed and load. Poor acceleration or response is difficult to troubleshoot since it can be caused by factors such as:

- Engine or pump related factors
- Driver technique
- Improper gearing
- Improper engine application
- Worn clutch or clutch linkage.

Engine related poor acceleration or response can be caused by several different factors such as:

- Failed boost sensor
- Excessive drain line restriction
- Throttle dead band.

See the performance tree Engine Acceleration or Response Poor for the proper procedures to locate and correct a poor acceleration or response complaint.

Driveability/Low Power - Customer Complaint Form

Customer Name/Company _____ Date _____

1. How did the problem occur? Suddenly _____ Gradually _____
2. At what hour/mileage did the problem begin? Hours _____ Miles _____ Since New _____

- After engine repair? Yes _____ No _____
- After equipment repair? Yes _____ No _____
- After change in equipment use? Yes _____ No _____
- After change in selectable programmable parameters? Yes _____ No _____
- If so, what was repaired and when? _____
- Does the vehicle also experience poor fuel economy? Yes _____ No _____

Answer questions 4 through 8 using selections (A through F) listed below. Circle the letter or letters that best describes the complaint.

- A. Compared to fleet
- B. Compared to competition
- C. Compared to previous engine
- D. Personal expectation
- E. Will **not** pull on hill
- F. Will **not** pull on flat.

A B C D E F

- Can the vehicle obtain the expected road speed? Yes _____ No _____
- What is desired speed? rpm/mph _____
- What is achieved speed? rpm/mph _____
- GVW _____

A B C D

- Is the vehicle able to pull the load? Yes _____ No _____

When?

- _____ In the hills
- _____ With a loaded trailer
- _____ On the flat
- _____ Other _____

If question 4 or 5 was answered no, fill out the Driveability/Low Power/Excessive Fuel Consumption Checklist and go to the Low Power performance tree. A B C D E F

- Is the vehicle slow to accelerate or respond? Yes _____ No _____
- From a stop? Yes _____ No _____
- After a shift? Yes _____ No _____ rpm _____
- Before a shift? Yes _____ No _____ rpm _____
- No shift? Yes _____ No _____ rpm _____

A B C D

- Does the vehicle hesitate after periods of long deceleration or coasting? Yes _____ No _____ rpm _____

If question 6 or 7 was answered Yes, fill out the Driveability/Low Power/ Excessive Fuel Consumption Checklist and go to the Poor Acceleration/Response performance tree.

A B C D E F

Additional Comments:

This form can be copied for convenience.

Driveability/Low Power/Excessive Fuel Consumption - Checklist

Vehicle/Equipment Specifications

Year, Type and Model: _____

Transmission (RT 14609, etc.): _____, Duty Cycle: _____

Rear Axle Ratio, No. of Axles: _____, Application: Industrial ____, Marine ____. Genset ____. Automotive ____

Typical GVW: _____, Engine Rating: _____

Trailer Type and Size: _____, Height: _____, Weight: _____

Tire Size (11R x 24.5, Low Profile, etc.) _____

Tire Type: Radial _____, Standard Tread _____, Extra Tread _____

Fan Type: Direct Drive _____, Viscous _____, Clutch _____

Power Steering: Yes _____ No _____

Air Conditioner: Yes _____ No _____

Air Shield: Yes _____ No _____

Freon Compressor: Yes _____ No _____

General Information:					
DO Number:			SC Number:		
Fuel Pump Code:			Fuel Pump Serial Number:		
Mileage:			Engine Serial Number:		
Date in Service:			Engine Model and Rating:		
Cruise Speed and rpm:			Rated Speed and rpm:		
Road Speed Governor:	Yes	No	Type:		
Engine Brake:	Yes	No	Type/Brand:		
Chassis and Other Related Items					
Tank Vents:	OK	Not OK	Obvious Fuel Leaks:	Yes	NO
Brake Drag:	OK	Not OK	Axel Alignment:	OK	Not OK
Altitude:	Ambient Temperature:				
Typical Terrian:	Flat	Hilly	Other		
Fuel Heater:	Percent Asphalt:	Percent Concrete:			
Additional Comments:					

This checklist can be copied for convenience.					

Fuel Consumption - General Information

The cause of excessive fuel consumption is hard to diagnose and correct because of the potential number of factors involved. Actual fuel consumption problems can be caused by any of the following factors:

- Engine factors
- Vehicle factors and specifications
- Environmental factors
- Driver technique and operating practices
- Fuel system factors
- Low power/driveability problems.

Before troubleshooting, it is important to determine the exact complaint. Is the complaint based on whether the problem is real or perceived, or does **not** meet driver expectations? The "Fuel Consumption - Customer Complaint Form" is a valuable list of questions that can be used to assist the service technician in determining the cause of the problem. Complete the form before troubleshooting the complaint. The following are some of the factors that **must** be considered when troubleshooting fuel consumption complaints.

1. Result of a Low Power/Driveability Problem: An operator will change driving style to compensate for a low power/driveability problem. Some things the driver is likely to do are, (a) shift to a higher engine rpm or (b) run on the droop curve in a lower gear instead of upshifting to drive at part throttle conditions. These changes in driving style will increase the amount of fuel used.

2. Driver Technique and Operating Practices: As a general rule, a 1 mph increase in road speed equals a 0.16 km [0.1 mpg] increase in fuel consumption. This means that increasing road speed from 50 to 60 mph will result in a loss of fuel mileage of 1.6 km [1 mpg].
3. Environmental and Seasonal Weather Changes: As a general rule, there can be as much as a 1.6 to 2.4 km [1 to 1.5 mpg] difference in fuel consumption depending on the season and the weather conditions.
4. Excessive Idling Time: Idling the engine can use from 0.11 to 0.33 liters [0.5 to 1.5 gal] per hour depending on the engine idle speed.
5. Truck Route and Terrain: East/west routes experience almost continual cross winds and head winds. Less fuel can be used on north/south routes where parts of the trip are **not only** warmer, but see less wind resistance.
6. Vehicle Aerodynamics: The largest single power requirement for a truck is the power needed to overcome air resistance. As a general rule, each 10 percent reduction in air resistance results in a 5 percent increase in mpg.
7. Rolling Resistance: Rolling resistance is the second largest consumer of power on a truck. The type of tire and tread design have a sizeable effect on fuel economy and performance. Changing from a bias ply to a low profile radial tire can reduce rolling resistance by about 36 percent.

Additional vehicle factors, vehicle specifications, and axle alignment can also affect fuel consumption.

Fuel Consumption - Customer Complaint Form

Customer Name/Company _____ Date _____

Answer the following questions. Some questions require making an X next to the appropriate answer.

1. What fuel mileage is expected? _____ Expected mpg
2. What are the expectations based on? Original mileage _____, Other units in fleet _____, Competitive engines _____ Previous engine owned _____, Expectations only _____, VE/VMS report _____
3. When did the problem occur? Since New _____, Suddenly _____, Gradually _____
4. Did the problem start after a repair? Yes _____ No _____ If so, what was repaired and when?

5. Is the vehicle also experiencing a Driveability problem (Low Power or Poor Acceleration/Response)? Yes _____ No _____

If answered Yes, fill out the Driveability/Low Power/Excessive Fuel Consumption Checklist and go to the Engine Power Output Low performance tree.

6. Is the problem seasonal? Yes _____ No _____
7. Weather conditions during fuel consumption check? Rain _____, Snow _____, Windy _____, Hot Temperatures _____, Cold Temperatures _____
8. How is the fuel mileage measured? Tank _____, Trip _____, Month _____, Year _____ Hubometer _____, Odometer _____
9. Are accurate records kept of fuel added on the road? Yes _____ No _____
10. Do routes vary between compared vehicles? Yes _____ No _____
11. Have routes changed for the engine being checked? Yes _____ No _____
12. What are the loads hauled, compared to comparison unit? GVW _____ Heavier _____, Lighter _____
13. What is the altitude during operation? Below 10,000 feet _____, Above 10,000 feet _____
14. How much of the time is the truck spent idling? Hours/day _____
15. Is the driver technique or operating practices affecting fuel economy?
 - High road speed: mph _____
 - Operate at rated speed or above: rpm _____
 - Incorrect shift rpm: Shift rpm _____, Torque Peak _____
 - Operate at a cruise speed: rpm _____
 - Believe compensating for low power: Yes _____ No _____

If after filling out this form it appears that the problem is **not** caused by vehicle factors, environmental factors, or driver technique, fill out the Driveability/Low Power/Excessive Fuel Consumption Checklist and go to the Fuel Consumption Excessive performance tree.

This form can be copied for convenience.

Oil Consumption

In addition to the information that follows, a service publication is available entitled Technical Overview of Oil Consumption, Bulletin No. 3379214.

Cummins Inc. defines "Acceptable Oil Usage" as outlined in the following table.

Acceptable Oil Usage									
Any Time During Coverage Period									
Engine Family	Hours per Quart	Hours per Liter	Hours per Imperial Quart	Miles Per Quart	Miles per Liter	Miles per Imperial Quart	KM per Quart	KM per Liter	KM per Imperial Liter
A	10.0	10.6	12.0	400	425	475	650	675	775
4B	10.0	10.6	12.0	400	425	475	650	675	775
6B	10.0	10.6	12.0	400	425	475	650	675	775
6C	10.0	10.6	12.0	400	425	475	650	675	775
V/VT-378	4.0	4.3	5.0	-	-	-	-	-	-
V/VT-504	4.0	4.3	5.0	250	265	310	400	425	485
V/VT-555	4.0	4.3	5.0	250	265	310	400	425	485
L Series	4.0	4.3	5.0	250	265	310	400	425	485
M Series	4.0	4.3	5.0	250	265	310	400	425	485
N Series	4.0	4.3	5.0	250	265	310	400	425	485
V/VT/VT A-903	4.0	4.3	5.0	250	265	310	400	425	485
KT/KTA-19	3.0	3.2	3.75	200	210	250	320	340	390
QSK23	1.7	1.8	2.0	-	-	-	-	-	-
V/VT/VT A28	2.0	2.1	2.5	-	-	-	-	-	-
KT/KTA38	1.5	1.6	1.8	-	-	-	-	-	-
KTA50	1.1	1.2	1.3	-	-	-	-	-	-
Acceptable Oil Usage (Transit Bus, Shuttle Bus and School Bus)									
Any Time During Coverage Period									
Engine Family	Hours per Quart	Hours per Liter	Hours per Imperial Quart	Miles per Quart	Miles per Liter	Miles per Imperial Quart	KM per Quart	KM per Liter	KM per Imperial Liter
B	10.0	10.6	12.0	200	210	240	320	340	385
C	8.0	8.5	10.0	150	160	180	240	255	290
L, M, N	4.0	4.3	5.0	100	105	120	160	170	195



Engine Lubricating Oil Consumption Report

Owner's Name	Date of Delivery			Engine Serial Number
	Month	Day	Year	
Address	Equipment Manufacturer			Engine Model & HP

City	State/Province	Equipment Serial Number	Fuel Pump Serial Number	
Engine Application (Describe)	Oil and Filter Change Interval		Complaint Originally Registered	
	Oil	Filters	Date	Mile/Hours/KM
Lubricating Oil Added				
Date Added Oil	Engine Operation Miles/Hours/Kilometers	Quarts - Liters Oil Added	Brand and Viscosity of Oil Used	
Start Test				
Last Mileage/Hours/Kilometers _____ Minus Start Mileage/Hours/Kilometers _____				
Equals Test Mileage/Hours/Kilometers _____ Divided By Oil Added _____				
Equals _____ Usage Rate _____				
Customer Signature	Cummins Dealer		Cummins Distributor	
Cummins Inc. Form 4755				



Oil Consumption Report	
Customer Name:	Dist/Dir:
Engine Model:	Mi/Km/Hr:
Engine Serial No.:	CPL No.:
Vehicle Make/Model:	Date:
Review of maintenance history: List any previous failures that could have had a detrimental effect on cylinder component life. Failures could include fuel, coolant, and/or foreign abrasives in the oil, second ring groove beat-out, filter plugging, etc.	
Lube Oil Used: Brand Viscosity Change Interval (mi/km/hr)	
Combination Oil Filter: Model Element Change Interval (mi/km/hr)	
Bypass Oil Filter: Model Element Change Interval (mi/km/hr)	
Full Flow Oil Filter: Model Element Change Interval (mi/km/hr)	
Air Cleaner: Make and Model Change Interval	
List any external engine leaks.	
Visually check for any internal leaks and list. Check turbocharger seals, valve guides, air compressor, etc.	
Had the fuel pump been tampered with? _____ What is maximum rail pressure readings? _____ If yes, the pump must be reset to factory specifications and the customer sent out to re-evaluate his oil consumption rate and the eligibility requirements must be met again.	
Warning: Governmental agencies have determined that used engine oil is toxic and carcinogenic. Avoid breathing, ingestion, and excessive contact. Drain and refill oil pan to check dipstick markings and notes findings.	
Only after above checks are completed, leaks corrected and proper documentation is completed, disassemble engine to determine cause for the failure and repair as required.	
State reason for oil consumption.	
Signed: _____	

Coolant Loss Pre-Troubleshooting Guide

Before troubleshooting, it is critical to know where the coolant is being lost. It is **not** always obvious where the missing coolant has gone.

Before troubleshooting, it is important to determine the exact complaint by interviewing the driver, looking at the service history and looking at the ECM information.

Driver Interview Questions

Driver's Name:

Engine Serial Number (ESN)

What is the complaint?

How is this engine used?

What are the load factors?

Where is the vehicle driven?

1. How is the radiator filled?
2. Is the radiator filled to the High or Low mark when the engine is cold (less than 60°C [140°F])?
3. What type of coolant is added?
4. Is there any coolant on the ground under your vehicle or engine?
5. Is there green or white streaks on the engine or near the coolant overflow hose?
6. Is there any specific condition (weather, altitude, or load) when indications of coolant loss is evident?
7. Does the engine ever overheat?
8. Does the warning light flash?
9. Under what conditions?
10. What temperature does the coolant run at normally?
11. Does the cooling fan operate correctly (fan on a 99°C [210°F])?
12. Is there any white smoke at operating temperature, or has anyone told you that white smoke is coming out of the exhaust?
13. Is oil analysis performed as part of the maintenance?
14. Are there elevated levels of sodium or potassium?
15. Has any increase in moisture condensation on the dipstick or oil fill cap, or moisture in the blowby been noticed?
16. Has a milky appearance been noticed in the lube oil that might indicate coolant is present.
17. What other comments did the driver/operator have that might help Cummins Inc. make the right repair?

Service History Review

Repeat cylinder head or cylinder head gasket repairs can indicate the problem is likely **not** the cylinder head or cylinder head gasket. Repeat problems can indicate a deeper problem in the engine. Keep this information in mind while going through the troubleshooting procedure.

Look at this engine's warranty claims history: who worked on engine last and what did technician do? How many miles/kilometers are on the engine? Has a cylinder head or cylinder head gasket been replaced before? At what engine hours were these repairs made?

ECM Data Review

Print out an INSITE™ electronic service tool Image Report from the ECM. Look for high temperature alarms or low coolant level alarms. Either indication confirms a complaint of losing coolant.

Are any fault codes logged in the Engine Protection Fault History?

ECM Fault Code 235 - Low Coolant (how many times)?

ECM Fault Code 151 - High Coolant Temperature (how many times)?

At this point, do you know where the coolant is going? If not and the coolant loss is not severe, suggest mounting a catch bottle on the radiator overflow tube to catch any overflow that can possibly be blowing out and becoming lost while at speed. Send the vehicle out to collect more data about where the coolant is or is not going. If the catch bottle has some coolant in it, refer back to the Coolant Loss External (out the overflow) interview questions.

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